

Supplementary Materials

Continuous Grip-Force Dynamics as an Online Biophysical Signal of Perceptual Strain

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Main manuscript: [manuscript.qmd](#) (HTML/PDF) · [manuscript-biorxiv.qmd](#) (bioRxiv preprint PDF)

1 Signal processing and feature extraction

Grip-force preprocessing followed the pipeline documented in the main Methods and in `analysis/01_ingest.qmd`:

1. Target-normalized error: $e(t) = (F(t) - F_{\text{target}})/F_{\text{target}}$
2. Linear detrending within each trial epoch
3. Fourth-order zero-phase Butterworth high-pass filter at 0.5 Hz (`sosfiltfilt`)
4. Welch PSD estimation (Hann window, 50% overlap, ~0.5 Hz resolution)
5. Band integration: visuomotor tracking (0.5–3.0 Hz) and physiological tremor (8.0–12.0 Hz)

Epochs spanned grip-cue onset through 1 s after probe onset. All spectral features were log-transformed before modeling.

2 Model specifications

Mixed-effects logistic regression. Perceptual accuracy was modeled as:

$$\text{logit } P(\text{correct}) = \beta_0 + \beta_1 \text{modality} + \sum_{k=1}^4 \beta_k \text{poly}_k(\text{difficulty}) + \beta_5 z(\text{tracking}) + \beta_6 z(\text{tremor}) + u_i$$

where $u_i \sim \mathcal{N}(0, \sigma^2)$ is a random intercept for participant i , and $z(\cdot)$ denotes within-condition standardized band power. Primary inference used high-grip trials only (40% MVC); low-grip models used the same structure.

Cross-validated machine learning. Random-forest and logistic classifiers were trained with subject-wise cross-validation (leave-one-subject-out) on high-grip trials. Feature sets compared:

- **Baseline:** stimulus difficulty level, task modality
- **DSP:** baseline features plus tracking- and tremor-band power

Performance was evaluated using area under the ROC curve (AUC).

Table 1: Cross-validated AUC for perceptual accuracy prediction (high-grip trials).

Feature set	Model	AUC	N trials
Baseline	Logistic	0.6170	7001
Baseline	RandomForest	0.8498	7001
DSP	Logistic	0.6279	6911
DSP	RandomForest	0.8553	6911

3 Sample coverage and attrition

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

Stage	Subject- tasks	Tri- als	% retain- able	Note
Oddball subject-tasks in behavioral v3	136	18,099	—	aud + vis; runs merged into trial counts
Excluded: aborted mid-task	13	478	—	beh_trials < 100

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

Stage	Subject- tasks	Tri- als	% retain- able	Note
Retainable subject-tasks	123	17,621	100.0%	individual missed trials already removed in v3
No grip files (entire subject $\times$ task)	1	128	0.7%	zero grip CSVs found; all behavioral trials in block excluded
Partial grip (behavioral > on-disk grip)	36	3,678	20.9%	some grip files exist but fewer trials than behavioral (often 1 run = 30 trials)
Grip trials on disk	122	14,430	81.9%	fatigue-break runs merged per session; retainable subject-tasks only
Feature extraction loss	95	671	3.8%	grip trials present but spectral features not extracted (short epochs, etc.)
Analysis set (trial features extracted)	122	13,759	78.1%	primary grip force modeling cohort

4 Additional figures and tables

Single-trial signal chain. Representative high-grip incorrect trial showing the transformation from raw force to normalized error, rolling variance, and Welch PSD with analysis bands marked.

Probe-locked spectrogram. Time–frequency power aligned to probe onset ($t = 0$), with normalized error in the upper panel.

Population spectral distributions.

Manipulation checks and individual differences.

Predictive modeling and brain–behavior.

Full fixed-effects table.

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

Model	Predictor	OR (95% CI)	p
Baseline (high grip)	Intercept	1.98 (1.60, 2.45)	< .001
Baseline (high grip)	Modality (visual vs. auditory)	1.70 (1.48, 1.96)	< .001
Baseline (high grip)	Stimulus difficulty (linear)	4.76 (3.86, 5.87)	< .001
Baseline (high grip)	Stimulus difficulty (quadratic)	30.19 (24.85, 36.68)	< .001

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

Model	Predictor	OR (95% CI)	p
Baseline (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	< .001
Baseline (high grip)	Stimulus difficulty (4th order)	1.75 (1.54, 2.00)	< .001
DSP-augmented (high grip)	Intercept	1.99 (1.61, 2.46)	< .001
DSP-augmented (high grip)	Modality (visual vs. auditory)	1.70 (1.47, 1.96)	< .001
DSP-augmented (high grip)	Stimulus difficulty (linear)	4.76 (3.86, 5.87)	< .001
DSP-augmented (high grip)	Stimulus difficulty (quadratic)	30.19 (24.85, 36.68)	< .001
DSP-augmented (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	< .001
DSP-augmented (high grip)	Stimulus difficulty (4th order)	1.75 (1.54, 2.00)	< .001
DSP-augmented (high grip)	Tracking power 0.5–3 Hz (z)	1.00 (0.93, 1.08)	0.965
DSP-augmented (high grip)	Tremor power 8–12 Hz (z)	0.96 (0.88, 1.04)	0.282

5 Reproducibility

Analysis notebooks live in `analysis/`:

Notebook	Purpose
<code>01_ingest.qmd</code>	Trial table construction and spectral feature extraction
<code>02_cognitive_motor_model.qmd</code>	Mixed-effects models and LC brain–behavior analyses
<code>03_hero_visuals.qmd</code>	Figure generation for manuscript and portfolio

Render from the repository root:

```
quarto render analysis/01_ingest.qmd
quarto render analysis/02_cognitive_motor_model.qmd
quarto render analysis/03_hero_visuals.qmd
```

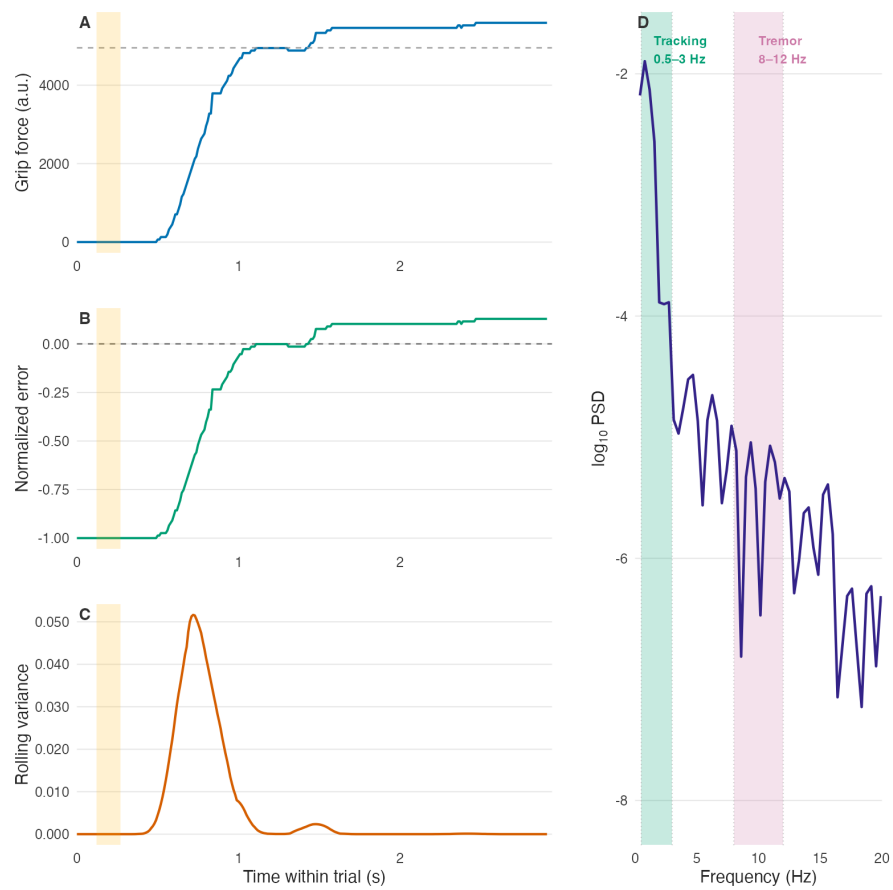


Figure 1: Single-trial signal chain from grip force to spectral phenotype.

The Cognitive Shock: Probe-Locked Motor Spectrogram

BAP171 · vis · session 2 · trial 8 · incorrect response

A = normalized error; B = short-time spectrum. Vertical gold line = oddball probe ($t = 0$). Grey shading = pre-probe baseline. Green band = tracking (0.5-

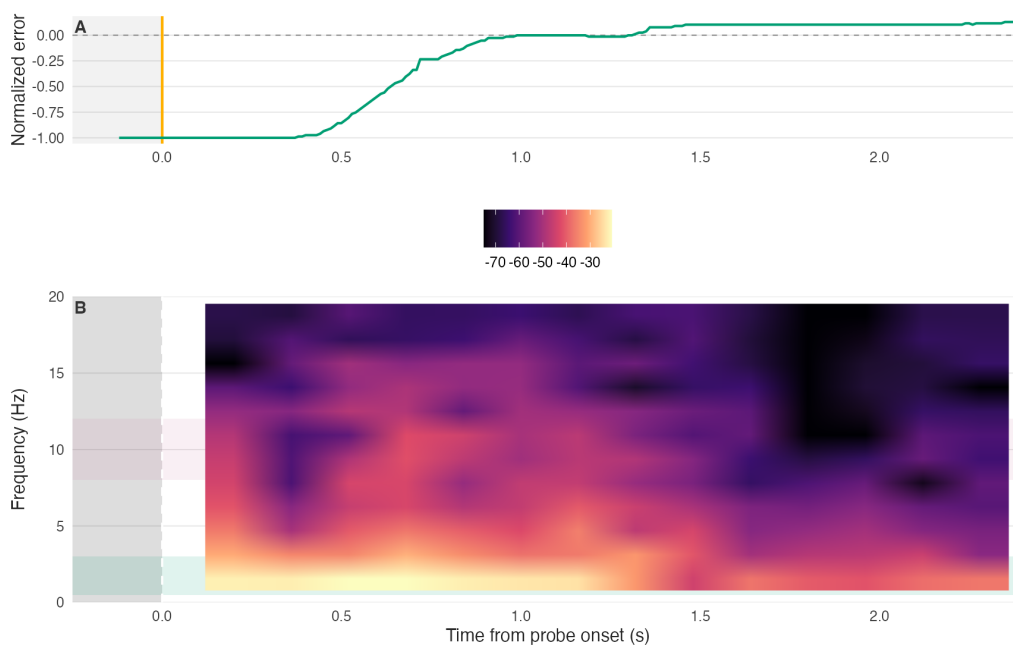


Figure 2: Probe-locked motor spectrogram aligned to oddball onset.

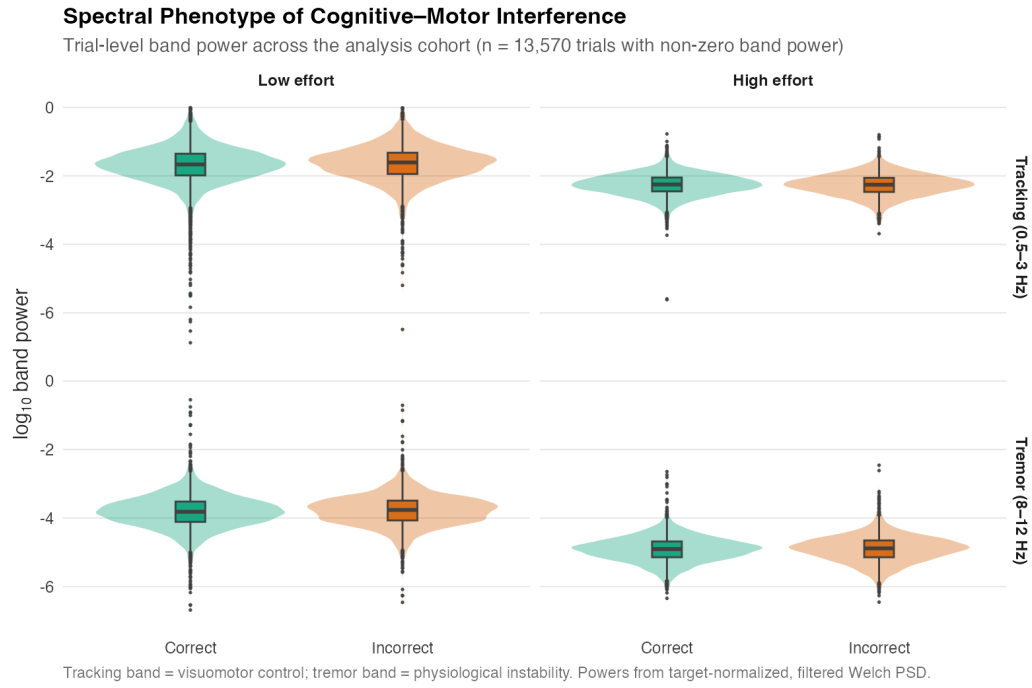


Figure 3: Trial-level log band-power by grip effort and perceptual outcome.

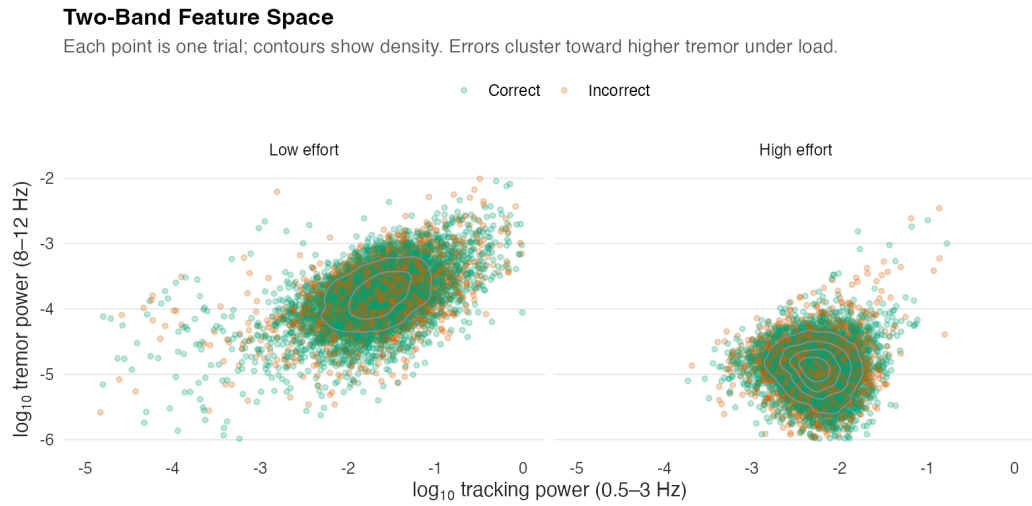


Figure 4: Two-band feature space (tracking vs. tremor), split by grip effort.

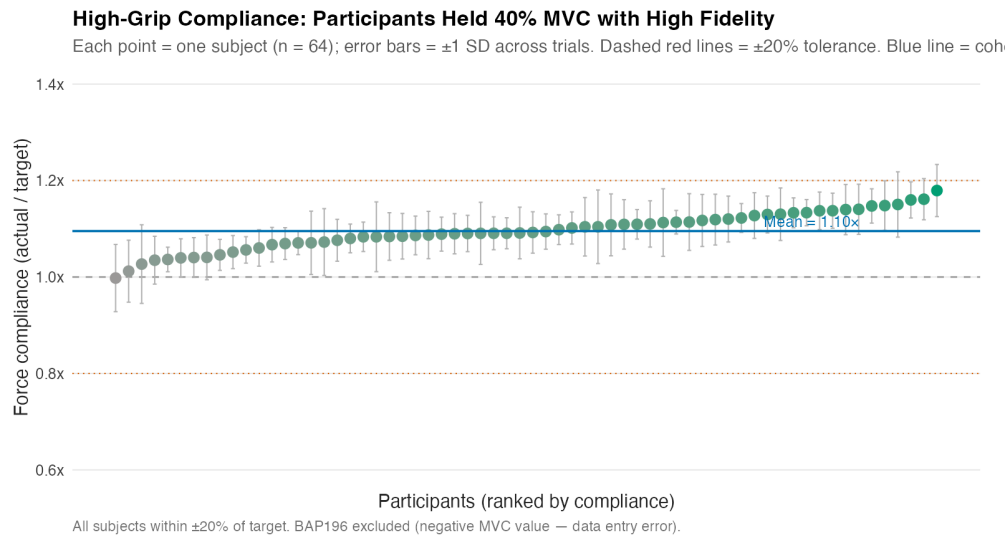


Figure 5: Per-subject high-grip force compliance (actual / target).

Substantial Individual Differences in Cognitive Load Cost

Each stem = one subject; positive = accuracy *drops* under high grip, negative = accuracy *improves*.
 ~70% of subjects show a detectable load cost; ~30% are facilitated, consistent with inverted-U arousal theory.

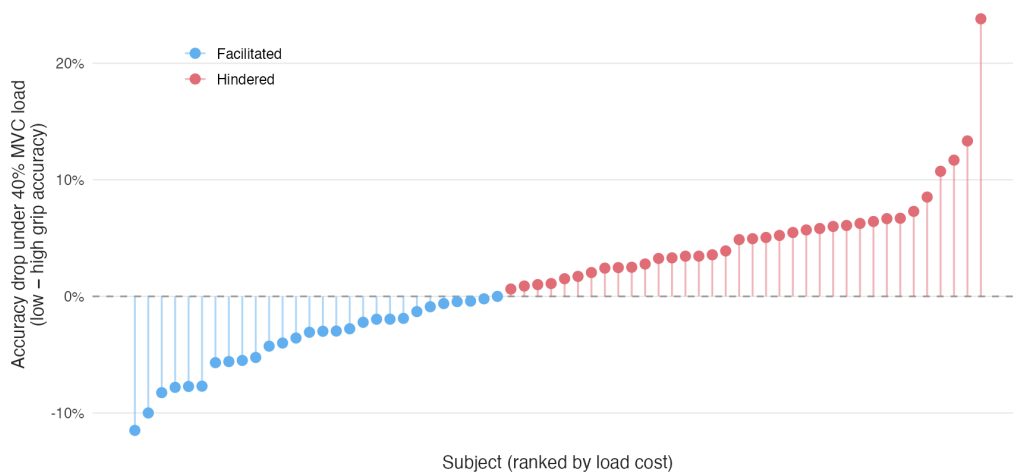


Figure 6: Subject-level accuracy drop under high versus low grip (caterpillar plot).

Physical Load Degrades Accuracy Without Buying Speed

Each line connects one subject's SAT point under low (blue) vs. high (red) grip. Diamond markers = grand mean $\pm$ SE. High grip shifts the cloud *down* (lower accuracy) with little systematic change in RT, ruling out a pure speed-accuracy swap.

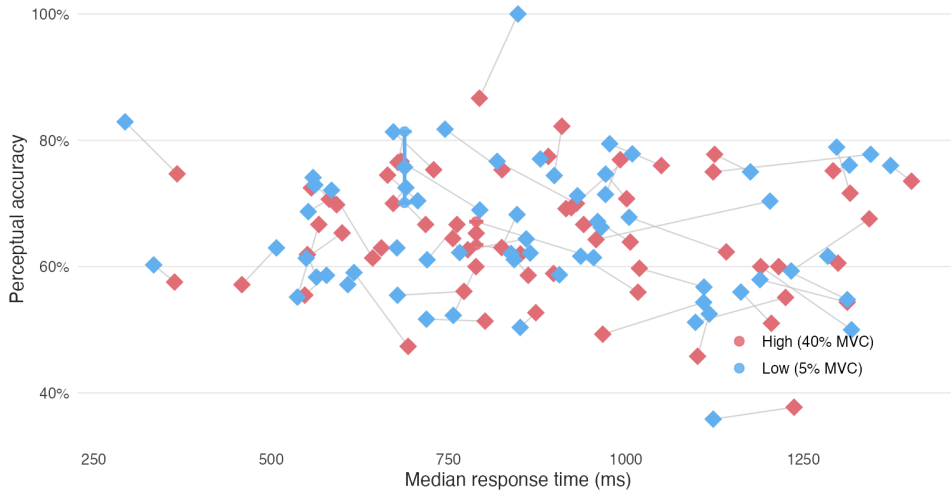


Figure 7: Speed-accuracy trade-off under low versus high grip.

Cross-validated prediction of perceptual accuracy

Random forest, subject-wise cross-validation, high-grip trials only

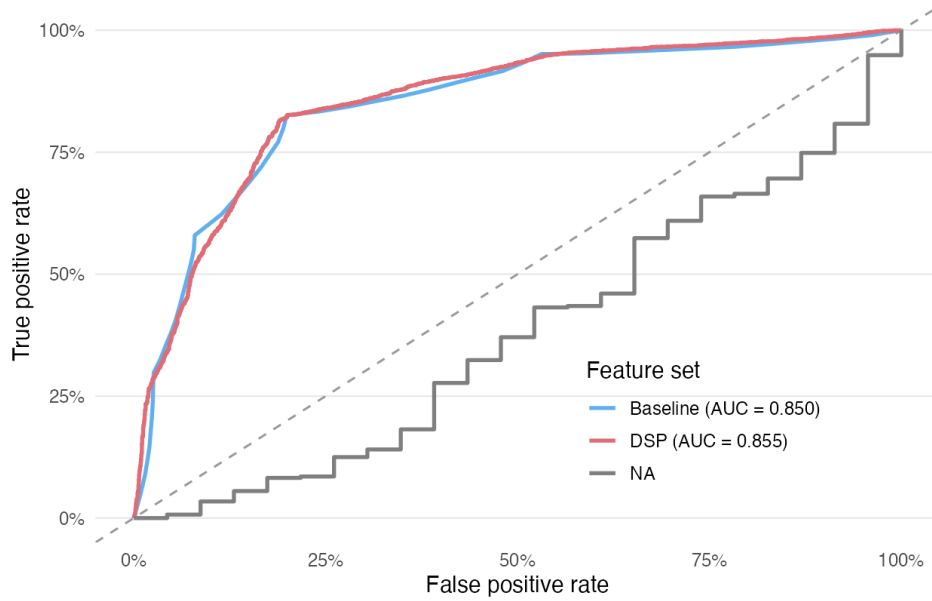


Figure 8: Cross-validated ROC curves (random forest, high-grip trials).

Exploratory LC microstructure and load-related motor reactivity
N = 55 with LC metrics and grip-force features

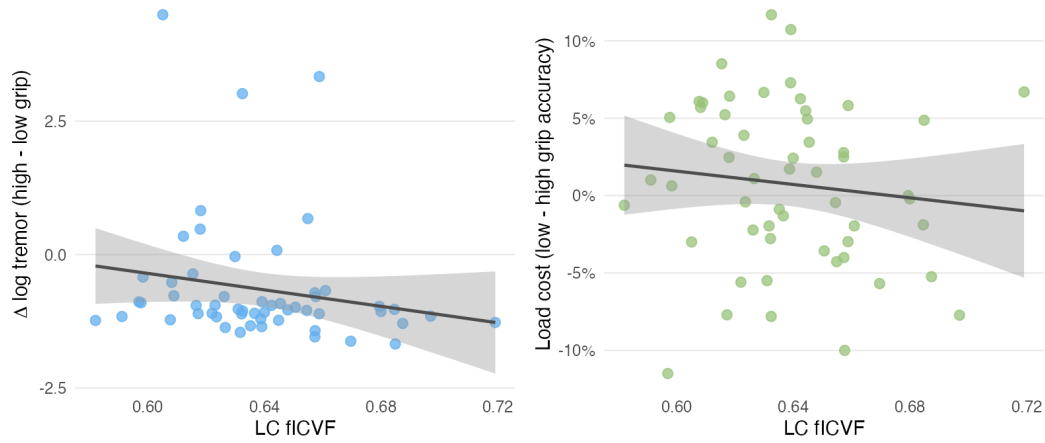


Figure 9: Exploratory LC fICVF associations with load-related tremor reactivity and perceptual load cost.

```
quarto render manuscript/manuscript.qmd
quarto render manuscript/manuscript-biorxiv.qmd --to biorxiv-typst
quarto render manuscript/supplementary.qmd
```

Local data paths are configured in `config/paths.local.yaml` (not tracked in git).